

Antimicrobial-resistant and extraintestinal pathogenic *Escherichia coli* in retail foods.



BACKGROUND: Extraintestinal *Escherichia coli* infections are associated with specialized extraintestinal pathogenic *E. coli* (ExPEC) strains and, increasingly, with antimicrobial resistance. The food supply may disseminate ExPEC and antimicrobial-resistant *E. coli*. METHODS: In a prospective survey of 1648 diverse food items from 10 retail markets in the Minneapolis-St. Paul area during 2001-2003, selective cultures and disk-diffusion assays for the isolation and characterization of antimicrobial-resistant *E. coli* and polymerase chain reaction-based assays and O serotyping to define ExPEC-associated traits were performed. RESULTS: *E. coli* contamination exhibited a prevalence gradient from miscellaneous foods (9%), through beef or pork (69%), to poultry (92%; $P < .001$). Among *E. coli*-positive samples, similar prevalence gradients were detected for antimicrobial resistance (27%, 85%, and 94% of samples, respectively; $P < .001$) and ExPEC contamination (4%, 19%, and 46%, respectively; $P < .001$). By multivariate analysis, beef or pork and poultry from natural-food stores exhibited reduced risks of *E. coli* contamination and antimicrobial resistance. Indirect evidence suggested on-farm selection of resistance. Four food-source ExPEC isolates (from pea pods, turkey parts, ground pork, and vegetable dip) closely resembled selected human clinical isolates by O antigen and genomic profile. CONCLUSIONS: Retail foods may be an important vehicle for community-wide dissemination of antimicrobial-resistant *E. coli* and ExPEC, which may represent a newly recognized group of medically significant foodborne pathogens.